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RESEARCH ARTICLE

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Growth performance, hematological and oxidative stress responses in Nile tilapia (*Oreochromis niloticus*) exposed to polypropylene microplastics

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Abstract

The present study investigated the effect of polypropylene (PP) microplastics on the growth performance, hematological profile and oxidative stress responses in the liver and brain tissues of *Oreochromis niloticus* juvenile. The microplastic exposure was done by offering microplastic-incorporated feed to the fish. The feed was prepared by mixing microplastics and the feed in three different ratios; Low (1.5 g/kg), Medium (3.0 g/kg), High (4.5 g/kg), and a control with no plastics in it. The trial was done for 7 weeks. A significant impact on the growth performance of the fish was observed in terms of body weight gain, specific growth rate, metabolic growth rate, protein efficiency ratio and energy retention. All the treatments had a significant difference in growth parameters compared to the control. The results of the hematological profile confirmed erythropoietic impairment in the fishes due to high microplastic exposure which led to an anemic condition in the fishes and mortality in the early juvenile stage. The antioxidant enzyme analysis also shows significant oxidative stress induced in liver and brain tissues in a dose-dependent manner. Exposure to PP microplastics significantly affected the growth, health and metabolism of *O. niloticus*. Further studies on genetic and cellular levels should be done to elucidate the wide-ranging effects of MPs on fish species.

KEYWORDS
growth performance, hematology, microplastics, *Oreochromis niloticus*, oxidative stress, polypropylene

1 | INTRODUCTION

The term 'Microplastics' is not new to the research community as well as the general public. But the sources and impacts of these tiny pollutants on the environment are not much known by the public community. Plastics have been already identified as a potential threat to the environment (Hodkovicova et al., 2022; Multisanti et al., 2022; Savuca et al., 2022). When these plastics degrade/break down into smaller particles through different environmental stresses such as sand, wave action, thermal degradation and photodegradation through UV exposure (Alimi et al., 2018), micro and nano plastics are formed. Plastic particles within a size range of 5–0.1 mm are generally considered Microplastics (MPs) (Haritha & Siddhuraju, 2022). The sources of these MPs are very vast and close to human activities or daily chores; hence it is very much easier and prone to disperse to different ecosystems including water, soil and atmosphere. Since we live in a setting where we frequently interact with all of these ecosystems, these MPs can enter the public quite quickly and in a variety of ways. However, the MPs become a more toxic pollutant when they return to humans than when they are dispersed. MPs get emitted from sources like cosmetics, personal care products, textiles and laundry, abrasion of household plastics, tire wear, fishing and aquaculture activities, etc. (Haritha & Siddhuraju, 2022). Both freshwater and marine water ecosystems serve as the main sink for most of the pollutants and thus MPs too (Burgos-Aceves et al., 2021a; Burgos-Aceves et al., 2022). All the above-mentioned sources finally end up with any type of water ecosystem either rivers, ponds, lakes, or oceans. Plastic debris of more than 10 million tons were released into the ocean each year (Urbanek et al., 2018) in which river emission accounts for around 2.4 million tons each year (Lebreton et al., 2017).

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Among the different ecosystems, freshwater ecosystems are found to be a major source, sink and also transporting media for MPs (Klein et al., 2018; Pastorino et al., 2023). These MPs can cause several harmful effects on the aquatic flora and fauna. Since freshwater fishes are largely consumed by humans and other animals throughout the world, the toxic effects of MPs on these fishes are of much concern. Primarily, the polluted fishes (MPs ingested) could act as an easy medium for the transfer of these toxic MPs to humans through the food chain (Qaiser et al., 2023). Various studies have proven that MPs could be detrimental to fish and other aquatic animals (Banaee et al., 2023; Impellitteri et al., 2023; Rakib et al., 2023). After ingestion, these MPs can affect fish growth and metabolism (Wu et al., 2023), block the digestive tract (Zicarelli et al., 2023), induce oxidative stress (Alberghini et al., 2022; Lombardo et al., 2022), cause intestinal inflammation, immune responses (Limonta et al., 2019), hypoactivity (Chen et al., 2017), potential neurotoxicity (Ding et al., 2018), reproductive toxicity (Sarasamma et al., 2020) etc. Accumulation of MPs in the gills, gut and body tissues of fishes and other aquatic organisms were already discussed in several studies (Aiguo et al., 2022; Burgos-Aceves et al., 2021b; Porretti et al., 2023; Yin et al., 2022). Now the toxicological and biochemical effects should be considered with high concern since fishes are a non-negligible part of the daily food habit of humans.

The test organism *Oreochromis niloticus* (Nile tilapia) is a common freshwater fish and also one of the world's most cultured and consumed fishes (Hamed et al., 2020). It is also used widely in toxicological studies (Hamed et al., 2021; Osman et al., 2018). Even if several studies have been conducted on MPs effects on fish, dietary exposure of MPs rather than water-borne exposure is less considered. Also, earlier studies on similar aspects don't give much importance to the monitoring of the growth performance of fish on MPs exposure which is a serious concern while coming to the quality and quantity of edible fish. Polypropylene is one of the most commonly used polymers for making bottles, plastic packaging, textiles, industry and machines. But only scanty data are available on the effect of PP in fish up to the best of our knowledge. Hence, the present study is proposed to examine the effects of polypropylene microplastics on the growth performance, hematological and biochemical conditions as well as oxidative stress responses of the organism.

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2 | MATERIALS AND METHODS

2.1 | Preparation of microplastics

Polypropylene beads were purchased from the commercial plastic industry at Coimbatore. Later, the polymer type was confirmed using ATR-FT/IR-4700 type A (Jasco, Japan). These were powdered by cryogenic grinding using a Ball mill/Mixer mill (MM 400, Retsch, Germany). Food-grade liquid nitrogen was used for freezing the plastic beads. A grinding jar of stainless-steel material of 50 mL capacity and a single grinding ball was used. An aliquot of plastic beads was filled in the grinding jar, which is then immersed in liquid nitrogen. A cryo-kit including insulated containers, tongs, safety glasses and safety gloves was used throughout the procedure. The embrittling time for the grinding jar is till the liquid nitrogen stops boiling. The jars were then transferred to the mixer mill and fitted properly. It is then agitated at 30 Hz grinding vibrational frequency for 1 min. The powdered plastics were then sieved through a stainless-steel sieve (5 mm pore size). Further the MPs particles which collected less than 5 mm were again sieved through a stainless-steel sieve (1 mm pore size) (Jayant Test Sieves, Jayant Scientific Industries, Mumbai, India). The final MPs particle size used for the study ranged from 0.2–1 mm which is later stored properly at ambient room temperature.

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2.2 | Preparation of microplastics incorporated fish feed

The powdered plastics were used to prepare microplastics incorporated fish feed. A commercial fish feed of 1.2 mm sized pellet (Growfin nutrition for fish, Growel feeds Pvt. Ltd., Andhra Pradesh, India-ISO 9001:2015 (International Organization for Standardization, Geneva, Switzerland) certified) was obtained from a fish feed store with required composition of mineral and vitamin premix. The experimental feed was prepared by mixing microplastics with the feed in three different ratios Low (1.5 g PP/kg of feed), Medium (3.0 g PP/kg of feed), High (4.5 g PP/kg of feed), and control without plastics. The proposed abovesaid concentration of microplastics used in the experimental study is based on the observations of microplastics concentration (quantitative and qualitative) present in the freshwater lakes during the field study (Haritha & Siddhuraju, 2023). A commercially available binder gel (Feedwale Powergel binder, Mumbai, India) was used to bind the powdered plastics to the feed. The feed with bonded plastics was then air-dried, labeled and stored in air-tight containers.

2.3 | Experimental setup and procedure

A fish feed trial was done for 7 weeks. Fingerlings (8–10 g) of *Oreochromis niloticus* were obtained from a local fish farm in Palakkad district, Kerala and kept in a tank for acclimatization. During the period the fish were fed with standard fish feed (control feed; 4% of body weight). After a week of acclimatization, the fish (12–15 g) were transferred to the treatment tanks. For each treatment, ie; control (PP-0gPP/kg), low (PP- 1.5PP/kg), medium (PP-3.0PP/kg) and high (PP-4.5PP/kg), three tanks for each treatment with eight fish in each tank respectively were used making a total of 12 tanks (four treatment × three tanks) and 96 fishes (12 tanks × eight fishes). The fish were starved one day before the start of the experiment and during the experimental period, they were fed at 4% of their body weight. The feeding was split into 4 equal rations per day. One-third of the water was changed every alternate day. The quality of water in the tanks was maintained at acceptable levels (Temperature 26.6 ± 1.1, Dissolved oxygen 6.5–8.0 mg/L, pH 6.8-7.8, Nitrite 0.07-0.1 mg/L, Nitrate 1–3 mg/L, Total hardness 156 ± 2.7 mg L^{-1} $CaCO_3$ and Total ammonia 0.1–0.3 mg/L) throughout the experiment in a natural photoperiod regimen.

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2.4 | Evaluation of growth parameters

The growth performance of fishes in control and treatments was assessed in terms of body weight gain percentage (BWG%), specific growth rate (SGR), feed conversion ratio (FCR), metabolic growth rate (MGR), protein productive value (PPV), Energy retention (ER) according to Dongmeza et al. (2006). In addition, the hepatosomatic index (HSI) and the intestine-somatic index were also calculated as:

$$\text{HSI} = (\text{Liver mass (g)} / \text{body mass (g)}) \times 100$$

$$\text{ISI} = (\text{Intestine mass (g)} / \text{body mass (g)}) \times 100$$

2.5 | Whole body proximate composition

The whole-body composition of the initial and final fishes was done according to AOAC (2019). Moisture, crude lipid, crude fiber and crude ash were done for initial and final fishes of control and treatments respectively. The energy content was estimated by multiplying crude protein, crude lipid and non-specific carbohydrate residues by the factors 16.7, 37.7, and 16.7, respectively (Siddhuraju et al., 1992).

2.6 | Microplastics extraction

After the 7 weeks of experimentation, the quantification of MPs was done in the fish intestine/gastrointestinal tract, gills and flesh. Using clean scissors and forceps the gills and gastrointestinal tracts of the fishes were removed. Later on, the dissected gills, gastrointestinal tract and flesh were digested for extracting microplastics using standard procedures (Karami et al., 2017; Pellini et al., 2018). The dissected gills, gastrointestinal tracts and flesh were soaked in 10% potassium hydroxide solution in glass bottles and incubated at 55°C for 48–72 h or till all the organic debris was digested. The digested solution was then filtered through a Glass microfiber filter (Grade-GF/A, Himedia Laboratories Pvt. Ltd., India) using a Vacuum filtration unit. Soon after filtration, filter papers were then dried and saved for microscopic examination.

After the 7 weeks of experimentation, the quantification of MPs was done in the fish intestine/gastrointestinal tract, gills and flesh. Using clean scissors and forceps the gills and gastrointestinal tracts of the fishes were removed.

2.7 | Microscopic examination

The microplastic particles extracted from the samples were identified using a trinocular microscope equipped with a camera (Weswox Optik Digital Research Microscope, India) at 4× and 40× magnification.

2.8 | Blood collection

After 7 weeks, at the end of the experiment blood samples were drawn from fish of each treatment for hematological analysis. For gentle handling and reducing stress during blood collection, fish were anaesthetized using clove oil: ethanol (1:9) mixture as an immersion bath at a dose of 100 μL per liter of water (Fernandes et al., 2016).

Blood was drawn via caudal vein puncture. The needle was inserted posterior to the anal fin, perpendicular to the ventral surface of the fish and blood was drawn carefully without changing the position of the syringe to avoid any mucus and water molecules entering the syringe. A single-use hypodermic syringe of 2 mL capacity was used. The collected blood samples were immediately transferred to purple colored vacutainer with EDTA as an anticoagulant, a red vacutainer with no

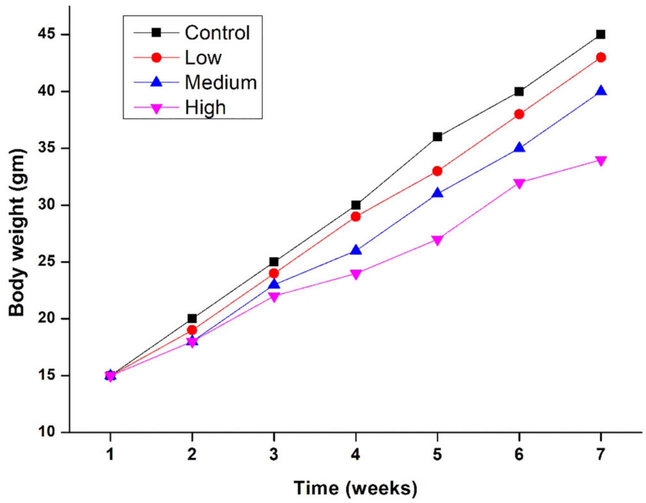

EXHIBIT 1 Graph showing growth performance of fish by weight. [Color figure can be viewed at wileyonlinelibrary.com]

additives and a grey vacutainer with sodium fluoride as a preservative for analysis of complete blood count, serum chemistry and blood glucose, respectively. These were then stored at 4°C till analysis.

2.9 | Hematological parameters

The hematological parameters RBC's, WBC's, Differential WBC's, Platelet count, total hemoglobin (Hb) and packed cell volume (PCV) were analyzed according to Mekkawy et al. (2011). The red cell indices, Mean Corpuscular Volume (MCV), Mean Corpuscular Hemoglobin (MCH), and Mean Corpuscular Hemoglobin Concentration (MCHC) were estimated according to the method of Iheanacho and Odo (2020).

2.10 | Biochemical parameters

Biochemical parameters such as glucose, cholesterol, total protein, albumin, globulin, A/G ratio, creatinine and uric acid were performed according to the method outlined by Hamed et al. (2019).

2.11 | Tissue collection for enzyme analysis

After blood sampling, the fish were further sacrificed for tissue sampling. The fish were dissected and the brain and liver were carefully collected which is rinsed in a physiological saline (0.9% sodium chloride solution) to remove any traces of blood.

2.12 | Antioxidant enzymes assay

The estimation of aspartate aminotransferase (AST) and alanine aminotransferase (ALT) was carried out by the method of Reitman and Frankel (1957) spectrophotometrically at 505 nm, which is expressed in unit IU/L. Whereas, Alkaline phosphatase (ALP) was estimated spectrophotometrically at 510 nm using the method of Kind and King (1972) in unit IU/L.

Superoxide dismutase (SOD) activity was estimated spectrophotometrically at 480 nm according to the method of Misra and Fridovich (1972). The activity is expressed in the unit IU/mg. Catalase activity was estimated according to the method of Beutler (1984) spectrophotometrically at 250 nm. Catalase activity is expressed in the unit of μmole of H_2O_2 consumed/min/mg protein. The activity of glutathione peroxidase (GPx) was estimated spectrophotometrically at 412 nm according to Beutler (1984) and the values were expressed in μmole of glutathione oxidized/min/mg protein. Lipid peroxidation assay was estimated according to Erdelmeier et al. (1998) at 525 nm using a spectrophotometer and expressed in nmoles of MDA mg^{-1} protein. The AChE activity was determined at 412 nm according to Ellman et al. (1961). The activity was expressed as μmole of acetylcholine hydrolyzed mg $protein^{-1}$ min^{-1}.

2.13 | Statistical analysis

All the values expressed were means of triplicate determination ± standard deviation. All the data were subjected to one-way analysis of variance (ANOVA), and the significance of differences between means was determined by Duncan's multiple-range test ($P < 0.05$) using SPSS (version 21, SPSS Inc., Waker drive Chicago, USA).

3 | RESULTS AND DISCUSSION

3.1 | General observations and growth performance

In most of the studies on MPs exposure in fishes, the path of exposure was found to be through the water. Ríos et al. (2022) observed an increase in the total ingestion of plastics during the presence of food, hence we preferred to offer microplastics via food rather than in water. The fish in the treatments were initially hesitant to actively feed on the pellets for the first two days of the trial. The mortality percentage (12.5%), one fish each at the end of the first week was observed in medium and high treatments. In which the deceased fish in the high treatment was found to have fin rot/tail rot disease based on the symptoms observed (discoloration at the end of the fin, fin fall off). Growth performance concerning body weight gain was observed (Exhibit 1). The control treatment had a linear weight gain when compared with the other three treatments. High treatment had moreover an irregular weight gain pattern, which confirms that the exposure to MPs had direct effects on weight gain and ultimately affected the growth performance of the fish. The ingestion of MPs through feed may cause blockages in the digestive tract and induce surfeit in the fish which can reduce feeding and hinder growth performance and also affect survival (Huang et al., 2020). Structural and functional deteriorations are also reported in gastrointestinal tracts of microplastics ingested fishes,

EXHIBIT 2 Growth performance, hepatosomatic and intestinal somatic indexes of *O. niloticus* exposed to PP MPs.

Parameter	Control	Low	Medium	High
IBW (g)	15.06 ± 0.15	14.93 ± 0.15	15.01 ± 0.11	15.02 ± 0.11
FBW (g)	44.94 ± 0.12^{a}	43.73 ± 0.45^{b}	41.8 ± 0.20^{c}	34.29 ± 4.50^{d}
BWG (%)	198.28 ± 2.31^{a}	193.12 ± 5.99^{b}	178.39 ± 7.9^{c}	128.09 ± 8.21^{d}
SGR (% day)	0.968 ± 0.006^{a}	0.953 ± 0.01^{b}	0.907 ± 0.02^{c}	0.726 ± 0.11^{d}
FI (mg fish^{-1} day^{-1})	245.02 ± 3.10^{a}	242.73 ± 4.40^{b}	229.92 ± 3.1^{c}	219.35 ± 5.51^{d}
FCR	2.00 ± 0.004^{b}	1.98 ± 0.04^{c}	2.01 ± 0.07^{b}	2.64 ± 0.60^{a}
PER	1.194 ± 0.002^{a}	1.153 ± 0.02^{b}	1.07 ± 0.04^{c}	0.77 ± 0.17^{d}
PPV (%)	24.68 ± 0.29^{a}	23.66 ± 0.45^{b}	23.23 ± 0.37^{b}	22.54 ± 0.28^{c}
ER (%)	16.44 ± 5.63^{a}	12.80 ± 5.08^{b}	12.16 ± 3.63^{c}	10.69 ± 4.6^{d}
MGR	7.59 ± 0.04^{b}	7.79 ± 0.15^{a}	7.36 ± 0.22^{c}	5.71 ± 1.0^{d}
ANLU (%)	11.81 ± 0.81^{a}	11.2 ± 0.7^{c}	11.4 ± 0.45^{b}	11.14 ± 0.33^{d}
Hepato-somatic index (HSI)	1.69 ± 0.08^{a}	1.67 ± 0.1^{b}	1.54 ± 0.06^{c}	1.43 ± 0.12^{d}
Intestine-somatic index (ISI)	5.11 ± 0.44^{d}	5.92 ± 0.42^{c}	6.10 ± 0.83^{b}	6.21 ± 1.39^{a}

Values of triplicate determinations (mean ± SD; $n = 8$) in the same column with different letters are significantly different ($P < 0.05$). Statistical program used- SPSS (version 21, SPSS Inc., Waker drive Chicago, USA).

which in turn could cause nutritional and growth issues in the organism (Wang et al., 2020).

The growth parameters, hepatosomatic and intestinal somatic index analyzed for control and treatments are given in Exhibit 2. Increasing the concentration of MPs in feed significantly reduced the growth performance of the fish. The treatment with a high concentration of MPs when compared to control, low and medium, was reported to have significantly ($P < 0.05$) lower growth performance. The final body weight (FBW), body weight gain (BWG) and specific growth rate (SGR) appeared to be significantly ($P < 0.05$) lower in all treatments compared to the control and the values were significantly different ($P < 0.05$) between the treatments also. It could be because the fish may use more energy to resist the hostile environment by increasing metabolic demands to maintain normal body functions, which may affect growth (Yin et al., 2018). Similarly, *Oreochromis urolepis* exposed to polyethylene MPs in water also showed a decline in the final weight, weight gain, total length and specific growth rate (Mbugani et al., 2022). Likewise, the dietary exposure of polyethylene MPs, decreased the weight gain of genetically improved farmed *O. niloticus* in a dose-dependent manner (Lu et al., 2022). However, in the present study the BWG and SGR were found to be below the levels observed in the above-mentioned study. This trend could be because of the difference in the feed used (ingredients and dietary composition of the fish feed), feed offered per day (rations/day), difference in the plastic polymer type used, size of the MPs particle used, experimental conditions, the age and strains of fish species, the micro-climatic conditions also etc. The poor growth resulted in a high feed conversion ratio, a low protein efficiency ratio, low protein productive value, low energy retention and apparent net lipid utilization. In general, the feed intake of fish was found to be significantly lesser ($P < 0.05$) in all the treatments compared to the control and further the lowest value registered in the high treatment. The same trend was also observed in the weight gain pattern of the respectively treated fishes. It could be due to the intake of MPs may cause a loss of appetite or a feeling like a satiated stomach (Bhuyan, 2022). Also, the presence of MPs particles in the feed pellets may make it uncomfortable for the fish to feed and poorly palatable too. Waterborne exposure to polystyrene MPs has reported to weaken the feeding activity of *Sebastes schlegelii* (Yin et al., 2018) and polypropylene microfibers negatively affected food consumption and a significant reduction in available energy for growth in *Carcinus maenas* (Watts et al., 2015). Further, due to the hydrophobic nature of PP MPs, it may hinder the digestion process and disrupt adequate absorption of micro and macronutrients post-digestion from the digestive tract (Wright et al., 2013). Wen et al. (2018) also found that PE microplastics adversely affected the digestion capacity of *Amazonian cichlid*. Interestingly, the experimental results revealed that all these growth parameters have a significant difference compared to the control in a dose-dependent manner in *O. niloticus*.

In all the treatments there was a significant reduction in HSI compared with the control. It decreased according to the increase in MPs concentration in the feed. This might be due to reduced fat absorption and subsequently reduced fat retention in the fish, particularly in the liver. The same trend of reduction is also seen in the fish body lipid. ISI increased significantly with an increase in MPs concentration. All the treatments had significantly higher ISI when compared with the control. It could be due to the intake of poorly/non-digestible MPs in the diet which increased the intestine mass. It was already observed that the fish of the above-mentioned treatment showed poor body growth.

3.2 | Whole body proximate composition

The initial and final whole-body composition of fishes are given in Exhibit 3. The crude protein content of the final fish significantly

EXHIBIT 3 Whole body composition of fishes before and after treatment.

Parameter	Before treatment	After treatment			
		Control	Low	Medium	High
Moisture (%)	77.3 ± 0.57	77.6 ± 0.57^{a}	76.23 ± 1.10^{b}	76.2 ± 0.5^{b}	77.8 ± 0.5^{a}
Crude protein (%)	14.18 ± 0.01	15.72 ± 0.04^{a}	15.7 ± 0.1^{a}	15.5 ± 0.09^{b}	15.42 ± 0.11^{c}
Crude lipid (%)	3.79 ± 0.01	4.59 ± 0.03^{a}	4.5 ± 0.02^{b}	4.51 ± 0.04^{b}	4.35 ± 0.03^{c}
Ash (%)	2.62 ± 0.02	2.80 ± 0.01^{a}	2.76 ± 0.01^{b}	2.63 ± 0.03^{c}	2.59 ± 0.01^{d}
Energy (kJ g^{-1})	4.07 ± 0.11	4.56 ± 0.089^{a}	4.45 ± 0.04^{b}	4.43 ± 0.01^{c}	4.39 ± 0.05^{d}

Values of triplicate determinations (mean ± SD; $n = 3$) in the same column with different letters are significantly different ($P < 0.05$). Statistical program used-SPSS (version 21, SPSS Inc., Waker drive Chicago, USA).

differed ($P < 0.05$) between the control and high treatment. Lipid, ash and energy were also significantly reduced when compared with the control. Only the moisture content was increased in treatment 'high', but not significantly differed from the control treatment.

Fish flesh/muscle, a major diet part is considered to provide 20% of animal protein consumed by humans (FAO, 2020). It is found that external factors like feed and environmental factors affect the nutritional quality of fish (Lai et al., 2021). The reduction in protein and ash of fishes in treatments than in control could be possibly due to the inadequate digestion and absorption of nutrients by fish. Similar results were observed in the study on HDPE exposure on *Perca flavescens* through feed, in which reduced protein and ash content were detected in higher concentration treatments (Lu et al., 2022). Moreover, a higher rate of protein degradation for energy and other physiological processes in the high and medium treatments may also be to account for the decline in protein. Hence significant effect on the nutritional quality of fish was observed after polypropylene MPs exposure.

3.3 | Microplastics quantification

Identification of microplastics was done through a trinocular microscope equipped with a camera. Microplastics were identified in all the treatments except the control, which confirms the accumulation of MPs in the fish. No recovery period was given in the experiment. More amount of microplastics was observed in High MPs exposure treatments than in the lower exposed treatments. No microplastics were identified in the control treatments. More amount of microplastics was observed in intestines than in gills in all the treatments, most likely because MPs were exposed through feed which is a direct route to the gastrointestinal tract. The MPs from the floating feed and feces may be responsible for the MPs observed in gills. No MPs were found in the flesh of fishes of any treatments. Microscopic images of extracted MPs from fish intestine of different treatments are shown in Exhibit 4.

3.4 | Hematology

The introduction of toxic substances into aquatic environments impacts the quality of water, causing shifts in the hematological characteristics of fish (Fazio et al., 2013). The results of the hematological profile of *Oreochromis niloticus* exposed to graded concentrations of PP are presented in Exhibit 5. PP exposure induced effects on most of the hematological parameters. RBC count increased upon exposure to higher concentrations whereas there was a decrease in Hb count. Metabolic changes associated with physical work, excitement, and stress responses upsurge the tissue oxygen requirements, RBCs ensure a sufficient supply of oxygen to tissues, hence large numbers of erythrocytes could be additionally employed and mobilized from the spleen. Even if there is enough count of RBCs, the Hb levels may be low due to malformed RBCs due to any impairment in erythropoiesis instigated due to any stress, here it may be the exposure to PP. This can be validated by the results of a recent study on PS nano plastics exposure on *Ctenopharyngodon idella*. In the study, there was no significant alteration in the erythrocytes count even after 20 days of exposure, but there were numerous abnormalities in the size and shape of erythrocytes such as notched, kidney-shaped and asymmetric nuclear constriction (Guimaraes et al., 2021). In most cases, erythropoietic impairment and disruption of erythrocyte maturation in aquatic animals occur when they are exposed to chemical stressors (Yaji et al., 2018). Similarly, Iheanacho and Odo (2020) observed an increase in RBC count and a decrease in Hb count in African catfish (*Clarias gariepinus*) on exposure to PVC. The reduction of Hb count may lead to anemic conditions in fishes, and in juvenile phases, it could lead to mortality of fishes. In the present study, there was mortality (12.5%) of fish in medium and high doses of treatment during the first week. A similar study by Hamed et al. (2019) on the same species also stated that the MPs caused anemia and perturbations in other haemato-biochemical parameters that could lead to the mortality of fish in the juvenile phase. The dose-dependent evident decrease of Packed Cell Volume (PCV) could be due to possible interruption of the hematopoietic process (Okoro et al., 2019) caused by PP exposure. According to Sopinka et al. (2016), the increase or decrease in erythrocytes and PCV could be part of the defense response to different stresses, especially environmental or toxicant-induced stress.

A decrease in WBC count on an increase in MPs concentration shows the inhibition of WBC production by PP MPs. These low WBC counts may lead to conditions like leucopoenia (Iheanacho & Odo, 2020), which makes the fish more prone to get diseases quickly since these WBCs play a defensive role in the immune system of organisms

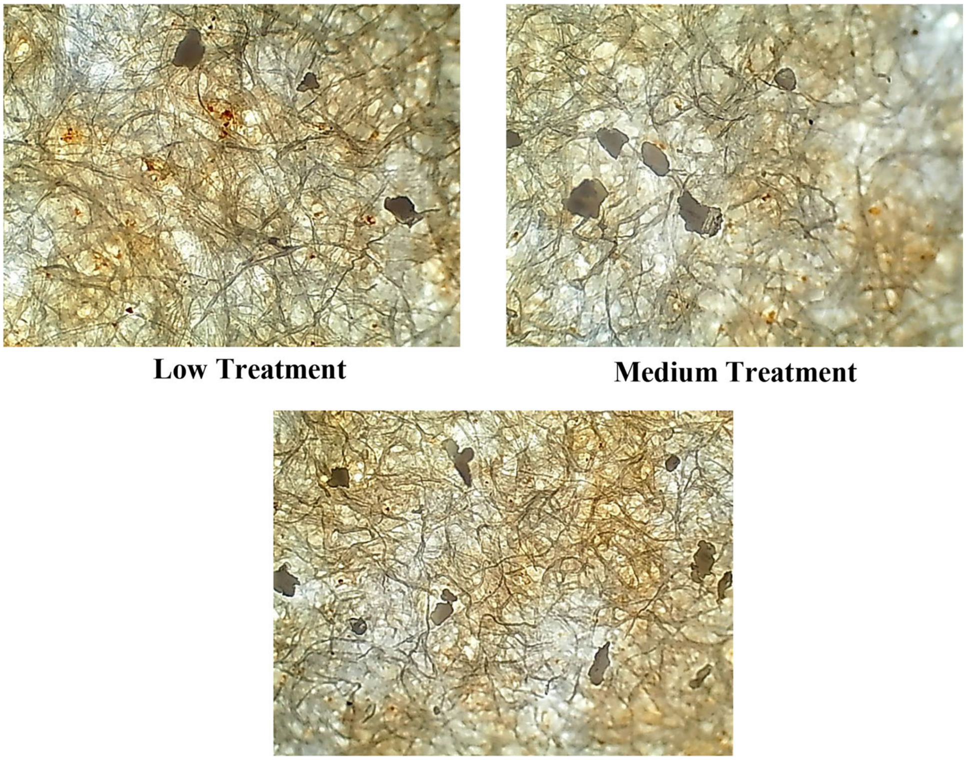

EXHIBIT 4 Microscopic images of extracted MPs from fish intestine of different treatments. [Color figure can be viewed at wileyonlinelibrary.com]

EXHIBIT 5 Effect of PP MPs exposure on hematological profile of *O. niloticus*.

Parameters	Control	Low	Medium	High
Total hemoglobin (Hb) (g/dL)	11.44 ± 0.1[a]	10.4 ± 0.1[b]	10.3 ± 0.17[b]	10.13 ± 0.15[b]
Packed cell volume (PCV) (%)	41.7 ± 0.3[a]	38.13 ± 0.1[b]	36.2 ± 0.26[c]	23.43 ± 0.20[d]
Total WBC count ($10^{-3}/\mu L$)	30.86 ± 0.3[a]	22.73 ± 0.1[b]	20.76 ± 0.25[c]	16.53 ± 0.20[d]
Polymorphs (%)	22.66 ± 0.5[d]	26.66 ± 0.5[c]	31 ± 1[b]	35.33 ± 0.57[a]
Lymphocytes (%)	65 ± 0.3[a]	60.33 ± 0.6[b]	56.33 ± 0.57[c]	55.33 ± 0.5 [c]
Monocytes (%)	6.33 ± 0.5[a]	7.33 ± 1.15[a]	6 ± 1[a]	3.66 ± 0.5[b]
Eosinophils (%)	6 ± 0.3[a]	5.66 ± 0.70[a]	6.66 ± 0.5[a]	6.33 ± 0.5[a]
Total RBC count ($10^{-6}/\mu L$)	3.4 ± 0.01[d]	4.04 ± 0.01[c]	4.11 ± 0.01[b]	4.68 ± 0.01[a]
MCV (μm^3)	90.33 ± 0.5[c]	88.33 ± 1.52[d]	92.33 ± 0.5[b]	94.66 ± 0.5[a]
MCH (pg/cell)	56.16 ± 0.05[a]	50.23 ± 0.05[c]	52.03 ± 0.15[b]	39.2 ± 0.1[d]
MCHC (g/100 mL)	37.13 ± 0.15[a]	23.2 ± 0.1[d]	25.16 ± 0.15[c]	33.1 ± 0.2 [b]
Platelet count (thousands/μL)	530.66 ± 0.5[d]	585.33 ± 1.15[c]	590.33 ± 1.52[b]	734.66 ± 1.5[a]

Values of triplicate determinations (mean ± SD; $n = 3$) in the same column with different letters are significantly different ($P < 0.05$); MCV-Mean corpuscular volume, MCH-Mean corpuscular hemoglobin, MCHC-Mean corpuscular hemoglobin concentration. Statistical program used-SPSS (version 21, SPSS Inc., Waker drive Chicago, USA).

(Jeney, 2017). The increase in MCV and decrease in MCHC values have been previously observed by Hamed et al. (2019) for early juveniles of the same species *O. niloticus* under MPs exposure, in addition in the present study, MCH also decreased. These red cell indices are useful in explaining the etiology of anemias (Sarma, 1990). Here MCV has been increased which shows that the size of red blood cells has increased and as stated the RBC count also simultaneously increased. But the decrease of MCHC and MCV shows that the produced RBCs don't have

EXHIBIT 6 Effect of PP MPs exposure on biochemical parameters of *O. niloticus*.

Parameters	Control	Low	Medium	High
Glucose (mg/dL)	79.16 ± 1.20[d]	89.36 ± 1.74[c]	101.06 ± 0.87[b]	103.4 ± 0.2[a]
Cholesterol (mg/dL)	143.26 ± 0.25[c]	182 ± 2.6[b]	199.96 ± 0.15[a]	201.3 ± 0.60[a]
Total protein (g/dL)	4.55 ± 0.02[d]	4.97 ± 0.01[c]	5.07 ± 0.01[b]	6.33 ± 0.01[a]
Albumin (g/dL)	1.95 ± 0.02[d]	1.99 ± 0.01[c]	2.36 ± 0.01[b]	2.57 ± 0.01[a]
Globulin (g/dL)	2.51 ± 0.01[d]	2.59 ± 0.01[c]	2.77 ± 0.01[b]	2.9 ± 0.01[a]
A/G Ratio (Ratio)	0.5 ± 0.01[c]	0.56 ± 0.05[c]	0.69 ± 0.01[b]	0.86 ± 0.05[a]
Creatinine (mg/dL)	0.60 ± 0.01[bc]	0.57 ± 0.02[c]	0.63 ± 0.03[b]	0.69 ± 0.01[a]
Uric acid (mg/dL)	13.6 ± 0.1[d]	14.66 ± 0.15[c]	15.5 ± 0.3[b]	16.4 ± 0.15[a]

Values of triplicate determinations (mean ± SD; $n = 3$) in the same column with different letters are significantly different ($P < 0.05$). Statistical program used-SPSS (version 21, SPSS Inc., Waker drive Chicago, USA).

enough amount of Hb or in other words, the produced RBCs were not healthy enough for their proposed functions. This confirms the effect of PP MPs in the erythropoiesis process and thereby inducing anemic conditions in the fish. As a whole, the disturbance in hematological parameters may be ascribed as a defense response against MPs toxicity (Osman et al., 2018).

3.5 | Biochemical parameters

All the biochemical parameters analyzed in the present study (creatinine, uric acid, glucose, cholesterol, total protein, albumin, globulin and A/G ratio) showed a significant increase with an increase in exposure concentration of MPs. The observed results are given in Exhibit 6. Following our results, similar changes in biochemical parameters were observed in the same species *O. niloticus* early juveniles (Hamed et al., 2019) and common goby (*Pomatoschistus microps*) after exposure to MPs (Oliveira et al., 2013). The glucose level of the body is maintained through homeostasis of glucose production and its storage as glycogen (Roda et al., 2020), any disturbance in this process could lead to variation in glucose levels in the body. MPs exposure to *Danio rerio* caused a significant increase in blood glucose levels, which was attributed to disturbed glucose metabolism in the liver (Zhao et al., 2020). Glycogen disintegration in hepatic tissue or deterioration in glucose absorption could be reasons for elevated glucose levels (Banaee et al., 2019). The liver also regulates lipid metabolism, such as cholesterol, and these MPs were also found to inhibit enzymes involved in lipid metabolism and alter the hormones related to it (Karami et al., 2016; Kim et al., 2021; Wu et al., 2024). This, in turn, alters cholesterol levels. Similar to our results, Brandts et al. (2021) observed a significant increase in blood cholesterol levels in *Sparus aurata* exposed to MPs (polymethylmethacrylate), which might lead to an energy crisis and nutritional problems in fish. Also, in contrast to our results, MPs exposure has decreased cholesterol and high-density lipoprotein levels in African catfish (Karami et al., 2016) and reduced cholesterol and triglyceride levels in the blood of common carp (Nematdoost Haghi & Banaee, 2017). The deterioration of hepatic and renal tissues could lead to an increment in albumin levels (Ramos-Barron et al., 2016), and an increase in globulin levels was regarded as an indicator of defense responses of the immune system (Hamed et al., 2019). Creatinine levels in the blood are a fairly reliable indicator of kidney function. As the kidneys become impaired for any reason, the blood creatinine levels increase because of a poor glomerular filtration rate. Also, as creatinine is a chemical waste molecule generated from muscle metabolism, its higher levels indicate the agile condition of the fish (Kulkarni & Pruthviraj, 2016).

3.6 | Antioxidant enzymes

The effect of PP MPs on liver and brain antioxidant enzyme activities of *O. niloticus* are presented in Exhibits 7 and 8, respectively. AST, ALP and ALT are different enzymes produced by the liver and these are considered reliable and sensitive bio-indicators for assessing damage primarily to the liver and other tissues under environmental and other stresses (Kim & Kang, 2016). The higher levels of these enzymes could also be an indicator of cell membrane damage (Banaee et al., 2014). In the study, the levels of these enzymes increased in the fish on an increase in exposure concentration of MPs indicating liver dysfunction and/or tissue damage.

The levels of antioxidant enzymes are analyzed to check if oxidative stress occurred in the test organism. These antioxidant enzymes are supposed to detoxify and eliminate the reactive oxygen species (ROS) and other chemicals which are normally produced during cellular metabolism (Ajima et al., 2017; Egea et al., 2017). Any imbalance in this elimination/detoxification process induces oxidative stress in the organism (Ajima et al., 2017) which could lead to cell and tissue damage (Song et al., 2006). The accumulation of MPs in the fish may increase ROS production which in turn stimulates antioxidant responses, indicating an increase in levels of antioxidant enzymes (Kim et al., 2021). SOD is considered a first-line antioxidant enzyme to deal with oxyradicals by accelerating the dismutation of superoxide to hydrogen peroxide, while CAT is a peroxisomal heme protein that catalysis the removal of hydrogen peroxide into water and oxygen formed by SOD. Thus, SOD and CAT are mutually supportive antioxidative enzymes (Filice et al., 2023; Kim et al., 2019, 2021). In the

EXHIBIT 7 Effect of PP MPs exposure on antioxidant enzymes in liver tissues of *O. niloticus*.

Parameters	Control	Low	Medium	High
Super oxide dismutase (U/mg protein)	0.168 ± 0.0005[c]	0.141 ± 0.001[d]	0.199 ± 0.001[b]	0.268 ± 0.0005[a]
Catalase (µmole/min/mg protein)	0.220 ± 0.001[d]	0.288 ± 0.001[c]	0.359 ± 0.0005[b]	0.489 ± 0.001[a]
Glutathione peroxidase (µmole/min/mg protein)	0.029 ± 0.0005[c]	0.029 ± 0.0005[c]	0.041 ± 0.0005[b]	0.070 ± 0.0005[a]
Lipid peroxide oxidation (nmole MDA/mg protein)	0.258 ± 0.001[d]	0.350 ± 0.0005[b]	0.341 ± 0.0005[c]	0.888 ± 0.001[a]
Acetylcholine esterase (µmole/mg protein/min)	0.419 ± 0.0005[c]	0.370 ± 0.0005[d]	0.469 ± 0.001[b]	0.699 ± 0.0005[a]
Aspartate transaminase (AST) (U/mg protein)	48.33 ± 1.15[c]	36.66 ± 0.57[d]	58.66 ± 0.57[b]	65 ± 1[a]
Alkaline phosphatase (ALP) (U/mg protein)	49.66 ± 0.57[d]	83.33 ± 0.57[b]	76.66 ± 0.57[c]	96.66 ± 0.57[a]
Alanine transaminase (ALT) (U/mg protein)	32.33 ± 0.57[d]	58.66 ± 0.57[c]	72.66 ± 0.57[b]	83.33 ± 0.57[a]

Values of triplicate determinations (mean ± SD; $n = 3$) in the same column with different letters are significantly different ($P < 0.05$). Statistical program used-SPSS (version 21, SPSS Inc., Waker drive Chicago, USA).

present study, SOD activity in the liver increased with a rise in exposure concentrations, but a significant increase to the control is seen only in the fishes exposed to high MPs concentration, which shows that only high concentration (4.5 g/kg) of MPs induced considerable SOD enzyme activity in the test organism. Similarly, in the brain also SOD levels were dependent on MPs exposure concentrations but a substantial increase in the control is seen only in the high-exposure treatment groups. An increase in SOD activity in the same organism *O. niloticus* following exposure to PS MPs was reported by Ding et al. (2018). Catalase enzyme levels in the liver and brain also followed a similar trend of SOD levels, in which a considerable increase is seen only in high exposure concentration when compared to the control, even if there was a steady increase in catalase levels on increase in MPs exposure concentrations. An elevation in levels of SOD and CAT enzymes was found in *Amazonian cichlid* following MPs (PS) exposure and the authors argued that this change in levels was induced by the antioxidant reaction to prevent oxidative damage caused by ROS production (Wen et al., 2018). Huang et al. (2020) also suggested that the MPs exposure enhances the SOD and CAT activity in *Poecilia reticulata*, due to increased ROS production and the trend of SOD and CAT levels was similar to the present study.

Glutathione peroxidase (GPx) is also a key enzyme that aids in the removal of hydrogen peroxide into alcohol and protects the organism against oxidative dysfunction (Ajima et al., 2017). Elevation in GPx activity is observed in the liver with an increase in exposure concentration with a significant increase in High MPs concentration compared to control, whereas in the brain only a slight increase was observed with not much significant increase even in High treatment. Exposure of PVC microparticles to African catfish also exhibited similar inhibition of GPx activity, which was suspected to be due to the increased generation of hydroperoxides owing to the toxic effects of PVC on the organ (Iheanacho & Odo, 2020). Ajima et al. (2017) stated LPO is a useful biomarker of oxidative damage and cell function due to oxidative stress in aquatic organisms. In the liver, LPO levels increased gradually and in high exposure concentration it increased significantly compared with control. And in the brain also a gradual increase in LPO levels was observed but was not much substantial and was comparable with the control. The rise in LPO levels indicates the attack of the ROS on lipid membranes of the respective organs thus causing oxidative dysfunction in the test

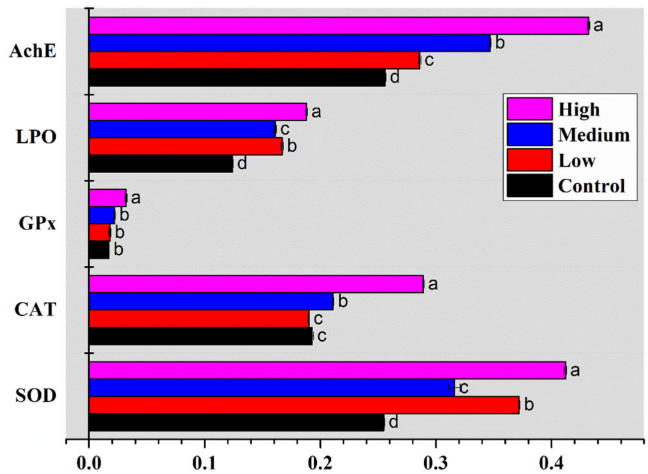

EXHIBIT 8 Graph showing effect of PP exposure on antioxidant enzymes in brain tissue of *O. niloticus*. [Color figure can be viewed at wileyonlinelibrary.com]

organism (Li et al., 2015). Hamed et al. (2020) reported elevated LPO levels in *O. niloticus* after MPs exposure. AChE is a notable biomarker for neurotoxic effects caused by toxicants and environmental stress on organisms (Iheanacho & Odo, 2020). AChE ensures proper functioning of the neuromuscular system via the deactivation of acetylcholine at the nerve endings (Ribeiro et al., 2017) and hence its activity levels denote the physiological state of the nervous system (Pala & Serdar, 2018). A significant increase in AChE activity is seen in brain tissues of the fishes whereas levels of AChE in the liver don't increase substantially. When comparing the antioxidant enzyme activity in the liver and brain, activities in the liver are more considerable/significant than in the brain, indicating more oxidative stress occurred associated with liver functioning.

4 | CONCLUSION

The study concludes that PP microplastics significantly affect the growth rate/performance and affected the health of *O. niloticus* by disturbing the antioxidant balance inducing oxidative stress, and inhibiting immune responses in the organism. Increased concentrations of MPs made significant effects on fish health. In the early stages of growth, PP microplastics caused anemic conditions which lead to fish mortality. Since *O. niloticus* is a commonly consumed fish, a significant reduction in the nutritional quality of the fish may adversely affect the food security and also the economic value of the fish. In a natural ecosystem, more adverse effects should be expected as the MPs there will be much more contaminated with several additives and other organic and inorganic contaminants.

Hence for future studies, MPs with combinations of other environmental toxicants or MPs recovered from natural ecosystems may be used for more realistic results. Most of the toxicological research is still being conducted at the individual and tissue levels, more cellular and genetic research is also needed for further and complete knowledge on MPs toxicity. Also, more research is carried out in the juvenile stage of the organism, studies should be conducted in other stages like larval and adult stages, and any gender-related differences in the effect should also be considered particularly during the reproduction stage. Moreover, the results observed in the present study could not be generalized for any type of polymer or organism. The effect may vary according to the type of polymer, size, exposure concentration, and organism, which should be studied in detail to make an environmentally relevant regulatory framework for MPs pollution or plastic concentration in natural aquatic ecosystems.

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CONFLICT OF INTEREST STATEMENT

The authors have no relevant financial or non-financial interests to disclose.

DATA AVAILABILITY STATEMENT

All data generated or analyzed during this study are included in this published article.

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